

LECTURE 27: MATH1061/7861
TOPICS: DEFINITIONS AND EXAMPLES OF
GROUPS
DATE: THURSDAY APRIL 30, 2026

Learning objectives:

- (1) Gain familiarity with groups, examples of groups, and their properties.

Fix a positive integer n .

We put a relation on $\mathbb{Z}$ as follows:

$$a \rho b \text{ if } n \mid (a - b).$$

(The same as requiring that a and b have
the same remainder modulo n).

We let $\mathbb{Z}_n = \{[0], [1], \dots, [n-1]\}$
denote the equivalence classes.

$$[0] = \{\dots, -n, 0, n, 2n, 3n, \dots\}$$

$$[1] = \{\dots, -2n+1, -n+1, 1, n+1, 2n+1, \dots\}$$

$\vdots$

What is a group?

A group is a set G together with an operation

$$\begin{aligned} f: G \times G &\longrightarrow G \\ (g, h) &\longmapsto g * h \end{aligned}$$

① Closure: image of f should be in G .

② Associative $(g * h) * k = g * (h * k)$.

(Note: Division or subtraction are not associative).

$$\underbrace{3 - (2 - 1)}_2 \stackrel{?}{=} \underbrace{(3 - 2)}_0 - 1$$

③ Identity: There is an element $e \in G$
 $e * g = g * e = g$

④ Inverses: Every $g \in G$ has an inverse
i.e. $\exists g^{-1} \in G$ s.t. $g * g^{-1} = e$
 $g^{-1} * g = e$

$(\mathbb{Z}, +)$ is a group

$(\mathbb{Z}, \times)$ is not group. Inverse of 4 is $\frac{1}{4} \notin \mathbb{Z}$.

~~$(\mathbb{Q}, \times)$~~ is a group
 $(\mathbb{Q} - \{0\}, \times)$

$(\mathbb{Z}_n, +)$ is a group.

$$x \in \{0, 1, \dots, n-1\} \quad -[x] = [-x] = [n-x]$$

$$\mathbb{Z}_7 \quad -[3] = [-3] = [4]$$

$(\mathbb{Z}_n - \{[0]\}, \times)$ $\leftarrow$ Sometimes a group
Sometimes not

 We now discuss this through two examples:
 $n=6$ and $n=7$.

Q1:

- (a) Write the full Cayley table of $\mathbb{Z}_7 - \{[0]\}$ under multiplication.

	1	2	3	4	5	6
1	1	2	3	4	5	6
2	2	4	6	1	3	5
3	3	6	2	5	1	4
4	4	1	5	2	6	3
5	5	3	1	6	4	2
6	6	5	4	3	2	1

Is this a group?

✓

(b) Is this a group? Is it abelian? Yes & Yes.

(Assume Associativity)

Closure: This holds as the table shows
if $a, b \in \mathbb{Z}_7 - \{[0]\}$ then
identity: 1 is identity so is ab .

Inverse: $1 \times 1 = 1$

$$2 \times 4 = 4 \times 2 = 1$$

$$3 \times 5 = 5 \times 3 = 1$$

$$6 \times 6 = 1$$

Q2: Do the same but now with $\mathbb{Z}_6 - \{[0]\}$. That is, write the Cayley table and determine whether it is a group.

	1	2	3	4	5
1	1	2	3	4	5
2	2	4	0	2	4
3	3	0	3	0	3
4	4	2	0	4	2
5	5	4	3	2	1

Not a group.

Closure fails: $2 \cdot 3 = 0 \notin \mathbb{Z}_6 - \{[0]\}$

Inverse fails: 2 has no inverse.